



Uncertainty and sensitivity analysis of cooling and heating loads for building energy planning

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ABSTRACT

It has become crucial to investigate the uncertainty and sensitivity of building loads (peak cooling load, peak heating load, annual cooling demand and annual heating demand) for meeting the risk assessment of building energy planning. Therefore, a new Monte Carlo (MC) method based on building performance simulation (BPS) is proposed to solve the problem of building loads forecasting at planning phase. Furthermore, the sensitivity of building loads is examined using two global sensitivity analysis (GSA) methods, including meta modeling method based on tree Gaussian process (TGP) and regression method based on standard regression coefficient (SRC). Finally, a case study of office building is conducted. The results show that the MC method constructed by the combination of R language platform and EnergyPlus software can generate models rapidly and simulate accurately building loads. Note that it is necessary to assess the stability of results as a function of sample size from uncertainty analysis in applying the MC method into building loads assessment. The TGP-based GSA method is applicable to identify and analyze key variables affecting building loads. It is recommended that at least two inherently different GSA methods should be applied to provide robust sensitivity results. Moreover, this study also provides insight on building energy planning and energy conservation design according to the results of uncertainty and sensitivity analysis for case study.

1. Introduction

Building energy planning is an effective way to reduce building energy consumption under the premise of ensuring the safety of building energy use [1–3]. As the first step of the planning, cooling/heating load calculation is very significant because it will affect the energy system configurations. At the planning stage, due to the lack of detailed architectural design scheme, the following two calculation methods are usually used to predict the building cooling/heating load. One is the area index method [4], another is deterministic building performance simulation (BPS) using professional platform (such as EnergyPlus, DOE-2, TRNSYS, and DeST). In order to ensure the energy system to have sufficient capacity, the common practice is to multiply the maximum load with a safety factor. Since the choice of an adequate safety factor is highly dependent on the designers' experience, the

resulting variation in the robustness of the system could be large. To avoid the responsibility of under-sizing, the safety factor is often larger than that actually needed. So the system is usually oversized, together with high capital cost and low operational efficiency. Therefore, it is necessary to carry out uncertainty analysis (UA) in the characteristics of cooling and heating loads at the planning stage to appropriately size equipment and reduce energy consumptions.

UA has been widely implemented in various areas of building energy analysis, including model calibration [5,6], load forecasting [7–9], impact and adaptation to climate change [10–12], sensitivity analysis (SA) [13,14], spatial analysis [15,16], and optimization [17–19]. Several studies have focused on uncertainty quantification of cooling and heating loads. Domínguez-Munoz et al. [20] proposed a probabilistic approach to consider the uncertainty of input factors influencing peak cooling load. Twenty uncertain factors are considered to calculate peak cooling load using a simple lumped heat-balance model. Sun et al.

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Nomenclature

BPS	building performance simulation
UA	uncertainty analysis
SA	sensitivity analysis
HVAC	heating ventilation and air conditioning
IDF	input data file
PDF	probability density function
CDF	cumulative distribution function
LHS	Latin hypercube sampling
GSA	global sensitivity analysis
TGP	treed Gaussian processes
MC	Monte Carlo
S_j	first order sensitivity indices
T_j	total effect sensitivity indices
X_j	input factors
X_{-j}	other factors excluding X_j
Y	model output
E	expectation
VAR	variance
ρ	Pearson correlation coefficient
SRC	standard regression coefficient
R^2	coefficient of determination
TMY	typical meteorological year

CSWD	Chinese standard weather data
AR	building aspect ratio
NF	number of floors
R_e	east window to wall ratio
R_w	west window to wall ratio
R_s	south window to wall ratio
R_n	north window to wall ratio
U_r	roof U-value
U_{wall}	wall U-value
U_g	ground U-value
U_{win}	window U-value
SHGC	solar heat gain coefficient
μ	mean value
σ	standard deviation

Subscripts

e	east
w	west
s	south
n	north
r	roof
wall	wall
g	ground
win	window

[21] further explored the heating ventilation and air conditioning (HVAC) sizing issues for both cooling and heating loads using UA, which can replace the safety factor commonly used in HVAC industry. This UA of cooling and heating loads takes into account actual observed weather data and provides rich probabilistic information on cooling and heating demands to support risk-based sizing. Lin Lin et al. [22] simulated and calculated the cooling load of a typical hub airport terminal on the basis of UA. The result shows that compared with the given parameters, more accurate average probability distribution and peak load distribution of terminal cooling load can be obtained under uncertainty parameters.

While UA and SA is closely related, they belong to two different disciplines. UA is focused on either assessing output uncertainty derived from input uncertainty or obtaining input uncertainty from measurement data, whereas SA concentrates on assessing the contributions of input factors for variations of system outputs. SA plays an important role in building energy analysis. It can be used to identify the key variables affecting building thermal performance from both energy simulation models and observational study. SA can be divided into two main categories, local and global [23]. A local sensitivity analysis (LSA) evaluates the response of the model to one local parameter. Global sensitivity analysis (GSA) explores the response of the model to changes in all input parameters, which are varied simultaneously [24]. GSA provides more information about the effect of varying model inputs but demands more computational effort. GSA can be further subdivided into four categories [25]: regression, screening, variance-based and meta-models. Many variables affecting cooling and heating loads are uncertain at the planning stage and the impact of these uncertainties is the reason for limiting the simulation accuracy.

The meta modelling method for GSA is a new one among these methods and consists of two steps: firstly construct a meta-model from detailed dynamic BPS; then implement a variance-based method using this meta-model [26]. The meta-models, also called surrogate models, are statistical models which approximate the objective function using the design of computer experiments [27]. The variance-based sensitivity method is to determine the contributions of individual factors to the variance of the output. This method is appropriate for non-additive non-linear model because of its model independence [24]. The combination of meta-models and variance-based method has several

advantages in comparison with other approaches for SA. Firstly, it needs less simulation runs compared to variance-based method because the disadvantage of variance-based SA is very time-consuming for engineering models and the meta-models have low computational cost. Secondly, it is suitable for complicated relationships between inputs and outputs due to the use of variance-based method, while the regression methods are appropriate for linear or monotonic models. Thirdly, it can quantify output variances due to every input, whereas the screening methods are usually only for qualitative purpose. Furthermore, the method can incorporate the uncertainty of both function outputs and integral estimation for sensitivity index, which are rarely used in most of SA methods [28]. Lastly, the method has a visual tool to show how the model output responds to every input factor [28], which can be very useful to show whether there are linear or non-linear relationships in the models.

Previous researches have not often focused on the uncertainty quantification of the four indicators of peak cooling load, peak heating load, annual cooling demand and annual heating demand at the planning stage. Moreover, the methods used in the previous studies are usually simplified models, which will bring additional model uncertainty to the uncertainty quantification of building loads. However, these building loads may have a critical impact on system design capacities for meeting the risk assessment of building energy planning. Therefore, the purpose of this paper is to construct a building loads forecasting method framework suitable for the lack of detailed building information in the building energy planning stage, and quantify the uncertainty of peak cooling load, peak heating load, annual cooling demand and annual heating demand using the proposed method framework to serve the practice of building energy planning.

Furthermore, previous studies have almost ignored the SA of peak cooling and heating loads to input parameters, especially the related elements of architectural design. Therefore, another aim of this paper is to implement GSA to identify and analyze key variables influencing peak cooling load, peak heating load, annual cooling demand and annual heating demand. More specifically, this paper examines the implications of the sensitivity of the different parameters for each result using meta modeling method based on TGP and regression method based on SRC. Moreover, this paper would investigate the difference between the

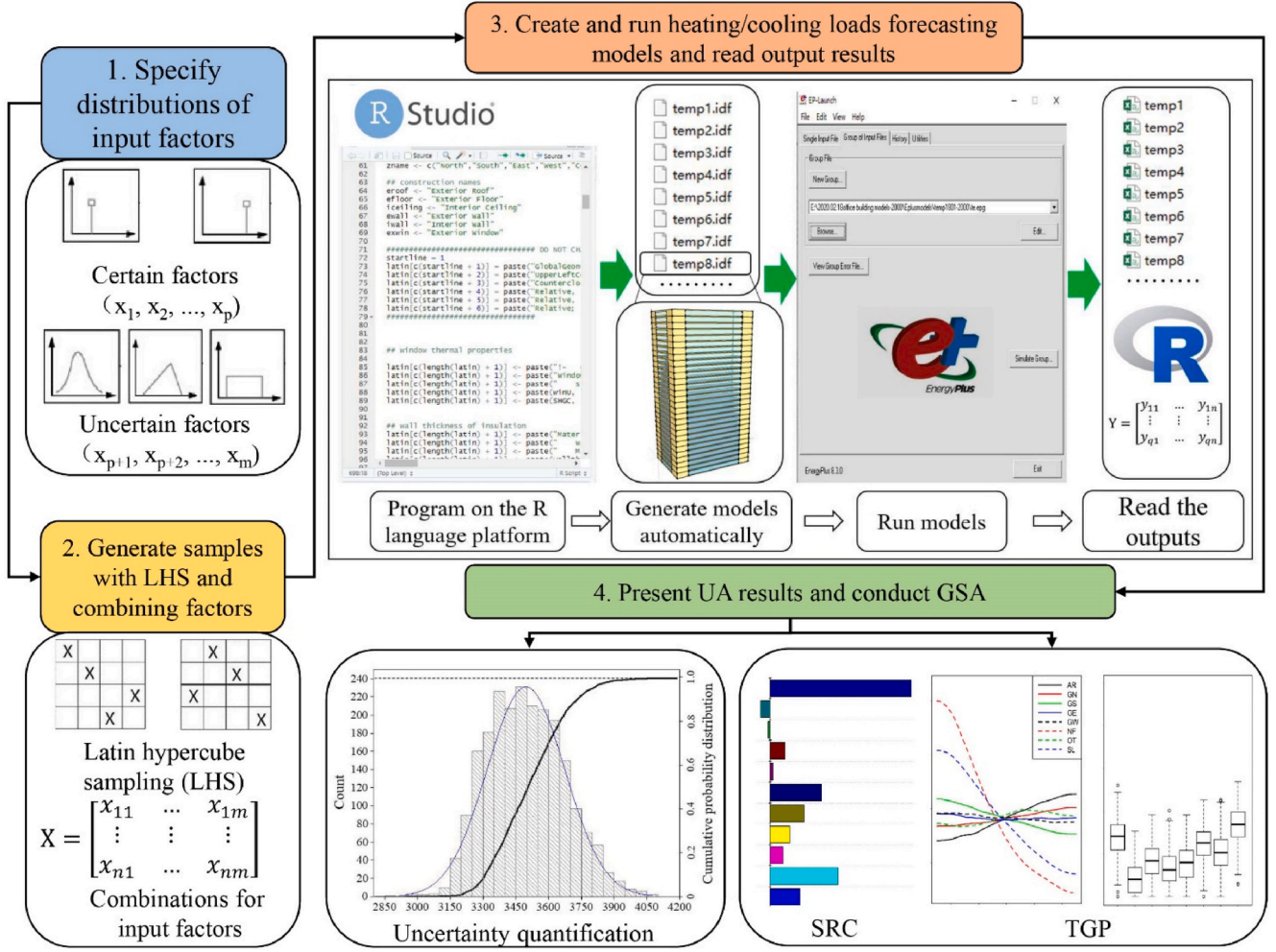


Fig. 1. Flow chat of the UA and GSA on the basis of MC method based on BPS.

effects of peak and annual analysis for cooling and heating demands of an office building, which can provide insight on energy demand from the perspectives of total energy demand and system capacity for building energy systems, respectively. This study would also be helpful to identify cooling and heating design best practices for office buildings to guide energy engineers during the building energy planning.

The paper consists of the following sections: Section 2 presents the proposed methodology including MC method based on BPS, UA method, meta modeling method based on TGP and regression method based on SRC. In Section 3, a case study of building energy model is described in detail. In Section 4, the sample size for UA and SA is analyzed selected, the uncertainty of peak cooling load, peak heating load, annual cooling demand and annual heating demand is quantified, and the main factors contributing to these four indicators are identified and analyzed using two GSA methods. Finally, the paper is closed with conclusions.

2. Methodology

The most important link for UA and GSA is to choose a method of uncertainty propagation. A MC method based on BPS constructed by the combination of R language platform and EnergyPlus software is proposed in this paper. This method has two main advantages: one is that a large number of models can be generated quickly by the combination of R language platform and EnergyPlus software and the interface of BPS model is easy to understand and operate; another is that compared with the uncertain input parameters, the mature EnergyPlus software used in this paper can ignore the uncertainty brought by the model, which can

provide more accurate data for UA and GSA of building loads. The specific implementation process of the UA and GSA on the basis of the proposed new MC method is illustrated in Fig. 1.

2.1. MC method based on BPS

2.1.1. Step 1 Specify distributions of input factors

The first step of MC method based on BPS is to specify distributions of input factors, including certain factors and uncertain factors. This step is both the most important and the most difficult task. For the certain factors, specific design values are determined according to standards and codes. As shown in Fig. 1, the factors from x_1 to x_p are deterministic, and they are set as point values. However, uncertainties will be characterized assigning probability distributions to the uncertain input factors, like the factors from x_{p+1} to x_m . These distributions characterize a degree of belief as to where the appropriate value of each variable is located. A probability distribution is expressed in terms of a probability density function (PDF), including Gaussian distribution, triangular distribution, uniform distribution, etc. It should be noted that Gaussian distributions need to be truncated where necessary to avoid unfeasible values (for instance, a negative value or zero for length-width ratio of building).

2.1.2. Step 2 Generate samples and combinations

The second step is to generate samples with Latin hypercube sampling (LHS) and combine the input factors. LHS used in this paper is a very popular method suitable for computationally demanding models

due to its stratification properties [15]. In short, LHS divides the range of every input variable into N segments (the specific sample size) with equal probability. The recommended length of the segments is dependent on the specified probability distribution shapes of input variables. One value is then randomly selected from each segment until all segments have one sample. An important feature in the LHS is that every hypercube partitioned by segments associated with input variables has the same number of samples. The general recommendation for a LHS sample size is ten times the number of variables in computer experiments [27]. However, this sample size may be inadequate for UA when complicated non-linear relationships between inputs and outputs exist in the cooling and heating loads models. Therefore, further research will be carried out in this paper to provide clear guidance on the sampling number suitable for the UA and GSA. LHS is implemented here using R software. Finally, a matrix X with m rows (sample size) and n columns (number of input factors) is generated by combining the certain and uncertain factors, as shown in Fig. 1.

2.1.3. Step 3 Create and run loads forecasting models

The third step is to automatically create building loads forecasting models using R software and run the models by EnergyPlus software. A large number of simulation runs are usually required in UA. Hence, it is necessary to automate the generation of simulation models with different input values, the simulation of models, and the extraction of the relevant data from simulation results using computer programming languages or specialized tools. In this paper, R software can use the input samples from the previous step to write the EnergyPlus models because the Input Data File (IDF) file required for EnergyPlus is an ASCII (i.e. text) file, which can be easily edited in the R language environment [29]. In order to reduce the time cost, EnergyPlus models are run on a multi-core computer and R platform is used again to automatically read the outputs from a large number of simulation runs in this paper.

2.2. UA method

As mentioned above, this paper adopts the MC method based on BPS. The MC simulation provides approximate solutions to a variety of mathematical problems by performing statistical sampling experiments on a computer [30]. It has the ability to use different distribution types for input factors and generate a probability distribution for each output [30].

In this paper, cooling and heating loads uncertainty propagation can be expressed as follows. Denote the loads forecasting model inputs (with p certain factors and $m-p$ uncertain factors) by $\vec{X}$, and the model outputs $\vec{Y} = f(\vec{X})$. The uncertainty of each parameter x_j ($j = p+1, \dots, m$) is quantified by its probability distribution. When the distributions of uncertain inputs have been defined, a sample is generated by randomly sampling values from each distribution using LHS. A matrix is then assembled that contains, for n model runs, the sampled values of each of the m input factors under examination, as shown in formula (1). The last step of the uncertainty propagation is to evaluate the model n times, once for each row of the sample (1), creating an input output map within the input space defined by the uncertainties. For each output, n possible results are obtained, as shown in formula (2).

$$\vec{X} = [X_1 \quad \dots \quad X_m] = \begin{bmatrix} x_{11} & \dots & x_{1m} \\ \vdots & \ddots & \vdots \\ x_{n1} & \dots & x_{nm} \end{bmatrix} \quad (1)$$

$$\vec{Y} = \begin{bmatrix} Y_1 \\ \vdots \\ Y_m \end{bmatrix} = \begin{bmatrix} f(x_{11}, x_{12}, \dots, x_{1m}) \\ \vdots \\ f(x_{n1}, x_{n2}, \dots, x_{nm}) \end{bmatrix} \quad (2)$$

Uncertainty quantification can be presented as numerical values or as visual graphs. Numerical indicators are descriptive statistical measures including mean, median, standard deviation and percentile etc.

Typical graphical methods include frequency distribution, Q-Q plot, probability density function (PDF), cumulative distribution function (CDF) and box plots etc. Frequency distribution shows that the data can be represented by some kind of distribution curve with a satisfactory fit. More specifically, this approach involves fitting a histogram based on output data with a suitable PDF. Q-Q plot demonstrates in how far the distribution matches the assumptions by means of a normality plot. Its purpose is to graphically assess whether the data follows a normal distribution. Box plots are used to show the quartile map. More valuable information can be provided based on PDF and CDF plots, such as confidence level and probability to exceed a threshold value. In this paper, the outputs of cooling and heating loads will be quantified by both numerical values and graphs.

2.3. TGP-based GSA method

The meta-modelling of GSA based on Treed Gaussian Process (TGP) will be used to identify and analyze the key variables affecting cooling and heating loads. TGP is one of the state-of-art machine learning methods, developed by Gramacy [31]. TGP couples stationary Gaussian process with treed partitioning, so it provides a flexible and powerful nonparametric and nonstationary regression model. And this method was further developed by Gramacy et al. [28] to be suitable for SA. The GSA method based on TGP can be implemented by R *tgpc* package, which was developed by Gramacy for R software [29]. Compared to other types of meta-model SA, the meta-model GSA based on TGP incorporates two types of variations for sensitivity index: uncertainty on function outputs; variability from integral estimation [31].

Two sensitivity indices can be provided by the meta-model GSA based on TGP: first order and total effects. The first order is to represent the main effect contribution of each input variable to the variance of the output. In contrast, the total effects is to represent the total contributions to the output variance due to each input factor by considering main effects and all higher-order effects. Therefore, the difference between the first order and total effects for one input variable indicates the interactions between this variable and other variables. The R *tgpc* package provides fully Bayesian non-stationary non-linear models (called treed Gaussian processes, TGP) and contains the sensitivity index calculation functions based on variance method. The functions allow for easy calculation of two very important sensitivity indices associated with each input: the first order for the j th input variable, as shown in formula (3),

$$S_j = \frac{\text{VAR}(E[Y | X_j])}{\text{VAR}(Y)} \quad (3)$$

and the total effects for input j , as shown in formula (4).

$$T_j = \frac{E(\text{VAR}[Y | X_{-j}])}{\text{VAR}(Y)} \quad (4)$$

where S_j is the 1st order sensitivity indices; T_j is the total effect sensitivity indices; X_j is the input factors; X_{-j} are the other factors excluding X_j ; Y is the model output; E is the expectation; VAR is the variance.

When using TGP method for sensitivity analysis, it is necessary to verify the correlation of input variables, and the accuracy of analysis results can be guaranteed when the input variables are irrelevant. Pearson correlation coefficient can be used to judge the correlation of input variables, as shown in formula (5).

$$\rho_{AB} = \frac{\text{cov}(A, B)}{\sigma_A \sigma_B} = \frac{E[(A - \mu_A)(B - \mu_B)]}{\sigma_A \sigma_B} \quad (5)$$

where ρ_{AB} is the correlation coefficient between variables A and B ; $\text{cov}(A, B)$ is covariance; σ_A and σ_B are the standard deviation of A and B , respectively; μ_A and μ_B are the mean of A and B respectively; E is the expected value. According to the Cauchy-Schwarz Inequality, ρ_{AB} is

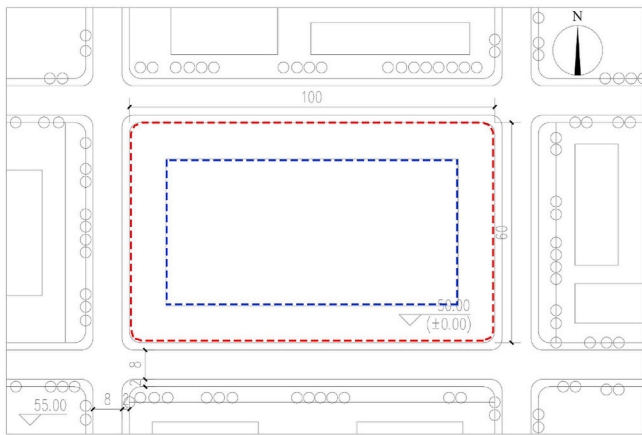


Fig. 2. Site plan.

between -1 and 1 . The absolute value represents the degree of correlation between the two variables. The greater the absolute value, the higher the degree of correlation.

2.4. SRC method

In order to ensure the reliability of sensitivity results, two or more methods are usually used to study the importance of influencing factors. Therefore, Standard regression coefficient (SRC) method will also be used to analyze the sensitivity of uncertain parameters in this paper [24]. The SRC can be calculated by formulas (6)–(9).

$$\frac{y - \bar{y}}{s} = \sum_{j=1}^N \frac{b_j \bar{y}}{s} \frac{x_j - \bar{x}_j}{s_j} \quad (6)$$

$$\bar{y} = \sum_{i=1}^N \frac{y_i}{N}, \quad \bar{x}_j = \sum_{i=1}^N \frac{x_{ij}}{N} \quad (7)$$

$$s = \left[\sum_{i=1}^N \frac{(y_i - \bar{y})^2}{N-1} \right]^{\frac{1}{2}}, \quad s_j = \left[\sum_{i=1}^N \frac{(x_{ij} - \bar{x}_j)^2}{N-1} \right]^{\frac{1}{2}} \quad (8)$$

$$SRC_j = b_j \frac{s_j}{s} \quad (9)$$

where, y is the model output; x_j ($j = 1, 2, \dots, n$) represents the j th input variable; b_j is the least squares estimation coefficient of each x_j ; N is the number of output variables; SRC_j is the standard regression coefficient of the j th input variable; s is the total standard deviation of the model output, and s_j is the standard deviation of the j th parameter. The value of SRC reflects the significance of the correlation between input variables and model output. The positive and negative signs of SRC represent the correlation direction between the input variable and the output variable.

The SRC method is suitable for the case where the input variables are independent of each other. Therefore, Pearson correlation coefficient can also be used for correlation analysis between variables before using SRC method. In addition, it is also necessary to judge whether there is a linear relationship between input variables and model output when using SRC method. Therefore, the coefficient of determination (R^2) is introduced to judge the fitting effect of linear regression. R^2 actually describes the interpretation degree of input variables to model output, and it can be calculated by formula (10).

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i - \bar{y})^2}{\sum_{i=1}^N (f_i - \bar{y})^2} \quad (10)$$

where, y_i is the real value; $\bar{y}$ is the average value; and f_i is the predicted value. The value of R^2 is between 0 and 1. When $R^2 > 0.7$, SRC method

Table 1

Planning information for the office building.

Parameter	Unit	Value
Site area	m ²	6000
Floor area	m ²	48000
Height between floors	m	4
Height limit	m	160
Plot ratio	—	8
Building density	—	0.4
Building orientation	°	0
Typical floor shape	—	Rectangle

Table 2

Information on some important certain inputs.

Parameter	Unit	Value	Ref.
Indoor set temperature in summer	°C	26	[33,34]
Indoor set temperature in winter	°C	20	[33,34]
Fresh air rate	m ³ /(h-person)	30	[33,34]
Occupant density	m ² /person	8	[33,34]
Equipment density	W/m ²	15	[33,34]
Lighting density	W/m ²	9	[33,34]

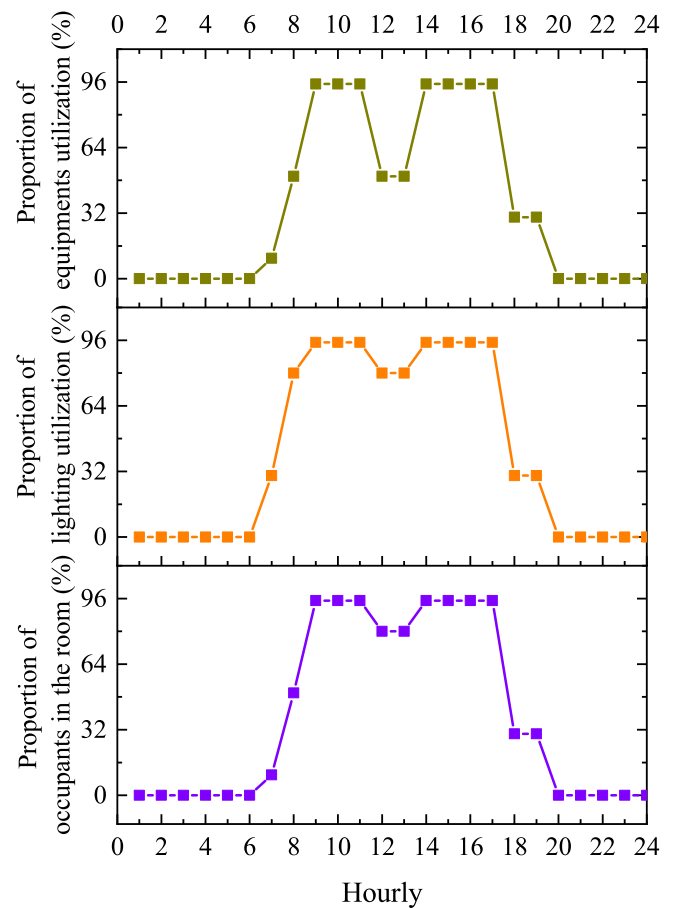


Fig. 3. Schedules for occupants, equipment and lighting [34].

Table 3
Information on the uncertain inputs.

Parameter	Short name	Unit	Distribution type	Distribution	Ranges	Ref.
Building aspect ratio	AR	–	Truncated normal	$N(1.2, 0.46^2)$	1–3	[15,25,33–37]
Number of floors	NF	–	Uniform	$U(20, 40)$	20–40	[15,25,33–37]
East window to wall ratio	R_e	–	Truncated normal	$N(0.46, 0.12^2)$	0.2–0.7	[15,25,33–37]
West window to wall ratio	R_w	–	Truncated normal	$N(0.46, 0.12^2)$	0.2–0.7	[15,25,33–37]
South window to wall ratio	R_s	–	Truncated normal	$N(0.55, 0.11^2)$	0.3–0.7	[15,25,33–37]
North window to wall ratio	R_n	–	Truncated normal	$N(0.55, 0.10^2)$	0.3–0.7	[15,25,33–37]
Roof U-value	U_r	$W/m^2 \cdot K$	Uniform	$U(0.1, 0.35)$	0.1–0.35	[15,25,33–37]
Wall U-value	U_{wall}	$W/m^2 \cdot K$	Uniform	$U(0.1, 0.45)$	0.1–0.45	[15,25,33–37]
Ground U-value	U_g	$W/m^2 \cdot K$	Uniform	$U(0.25, 0.45)$	0.25–0.45	[15,25,33–37]
Window U-value	U_{win}	$W/m^2 \cdot K$	Uniform	$U(1.0, 2.0)$	1.0–2.0	[15,25,33–37]
Solar heat gain coefficient	SHGC	–	Uniform	$U(0.15, 0.40)$	0.15–0.4	[15,25,33–37]

Note: For the uniform distribution $U(a, b)$, a and b are the lower limit and upper limit; for the normal distribution $N(c, d^2)$, c is the mean and d is the standard deviation.

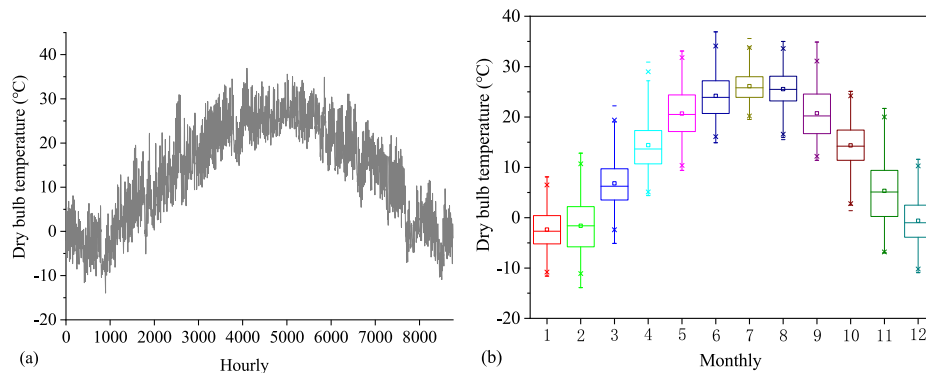


Fig. 4. Dry bulb temperature of typical meteorological year in Tianjin: (a) Hourly, (b) Monthly.

can be used to qualitatively evaluate the importance of input parameters in sensitivity analysis [32].

3. A case study of building loads model

3.1. Building conditions

As shown in Fig. 2, it is known that an office building will be built on the planning site (blue dashed box) in Tianjin, China. The plot ratio of the planned office building is 8, the building density is no more than 0.4, and the building height is limited to no more than 160 m. Other detailed planning information are shown in Table 1.

3.2. Model parameters

The office building loads model comprises several input factors that can be divided into two groups: certain factors and uncertain factors. The subgroup of certain factors includes those factors whose values can be determined by standards and codes [33,34]. For example, the indoor air temperatures in summer and winter are set to 26 °C and 18 °C respectively. The details of other important certain inputs of the model are shown in Table 2. Furthermore, detailed hourly schedules for occupants, equipment and lighting are also derived from a database of standards and codes [34], as shown in Fig. 3.

In this paper, eleven parameters related to architecture are selected as the uncertain input factors of the model. As shown in Table 3, these distributions have been set based on the survey data, theoretical considerations, literatures, standards and codes. Among them, building aspect ratio (AR), east window to wall ratio (R_e), west window to wall ratio (R_w), south window to wall ratio (R_s) and window to wall ratio (R_n) obey the truncated normal distributions, and the number of floors (NF), roof U-value (U_r), ground U-value (U_g), wall U-value (U_{wall}), window U-value (U_{win}), and solar heat gain coefficient (SHGC) follow the uniform distributions.

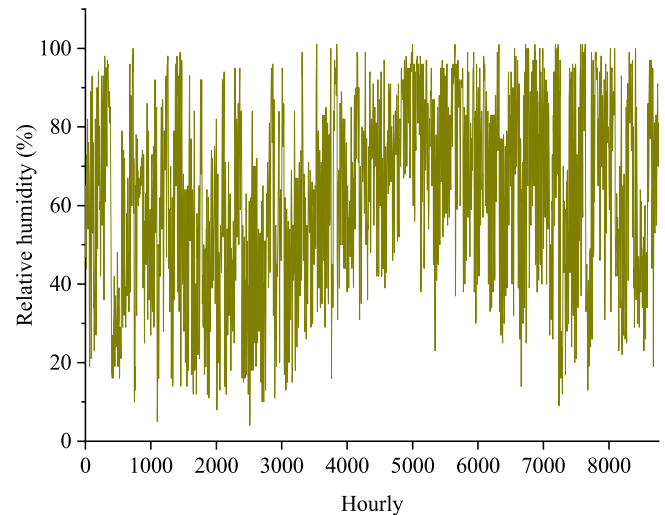


Fig. 5. Hourly relative humidity of typical meteorological year in Tianjin.

3.3. Climate information

The weather data of design day (winter design day and summer design day) and typical meteorological year (TMY) for Tianjin used in this paper is downloaded from EnergyPlus website. The weather data of Tianjin comes from Chinese standard weather data (CSWD), which is jointly prepared by China Meteorological Administration and Tsinghua University. According to the China's building thermal design zoning, Tianjin belongs to the cold area. The coldest month is January and the hottest month is July. The average temperatures of the coldest month and the hottest month are -3.5 °C and 27.0 °C respectively. Compared with other seasons, the relative humidity in summer is higher. The

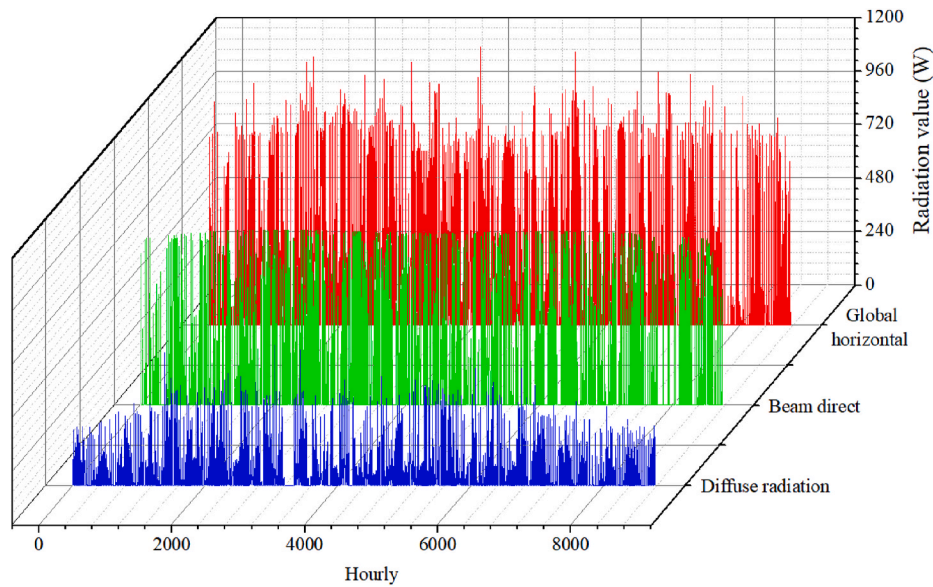


Fig. 6. Hourly solar radiation of typical meteorological year in Tianjin.

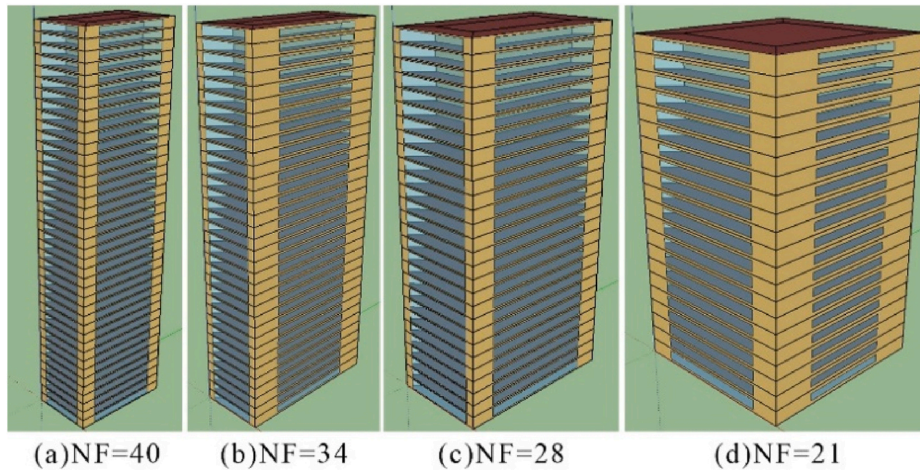


Fig. 7. Building loads models with different parameters.

relative humidity fluctuates greatly throughout the year. More detailed weather data of Tianjin are shown in Fig. 4, Fig. 5, and Fig. 6, including dry bulb temperature (hourly and monthly), hourly relative humidity and hourly solar radiation (global horizontal, beam direct and diffuse radiation) of TMY in Tianjin.

3.4. Model generation

After defining all the parameters required by the building loads model, the LHS method is used to sample eleven uncertain parameters. Then, a complete reference model of high-rise office building is established in the form of code on the R language platform, in which the uncertain parameters are defined as variables and other deterministic parameters are set as fixed values. Finally, the code is used to iterate the uncertain parameter values in each sample into the reference model, and the IDF format model recognized by EnergyPlus software is automatically generated. As shown in Fig. 7, four building loads models with different parameter values are automatically generated.

4. Results and discussion

4.1. Sample size analysis

Before conducting the UA and GSA, it is necessary to select the appropriate sample size to make the simulation results stable. The time cost and simulation accuracy caused by sample size is an inherent contradiction, so the determination of sample size needs to compromise these two aspects. In this paper, four percentiles (25th, 50th, 75th, and 95th) of peak cooling load, annual cooling demand, peak heating load and annual heating demand are proposed as monitoring values to determine the sample size. Starting with 55 samples, gradually increase the number of samples, and observe the fluctuation of simulation results. Each group of samples generated a considerable number of prediction models, so a total of 45210 models are generated in this study.

As can be seen from Fig. 8, the peak heating load and annual heating demand converge faster than the peak cooling load and annual cooling demand. However, under the same sample size after convergence, the relative error of cooling loads is smaller than that of heating loads. The percentiles of peak cooling load, peak heating load, annual cooling demand and annual heating demand become approximately stable after

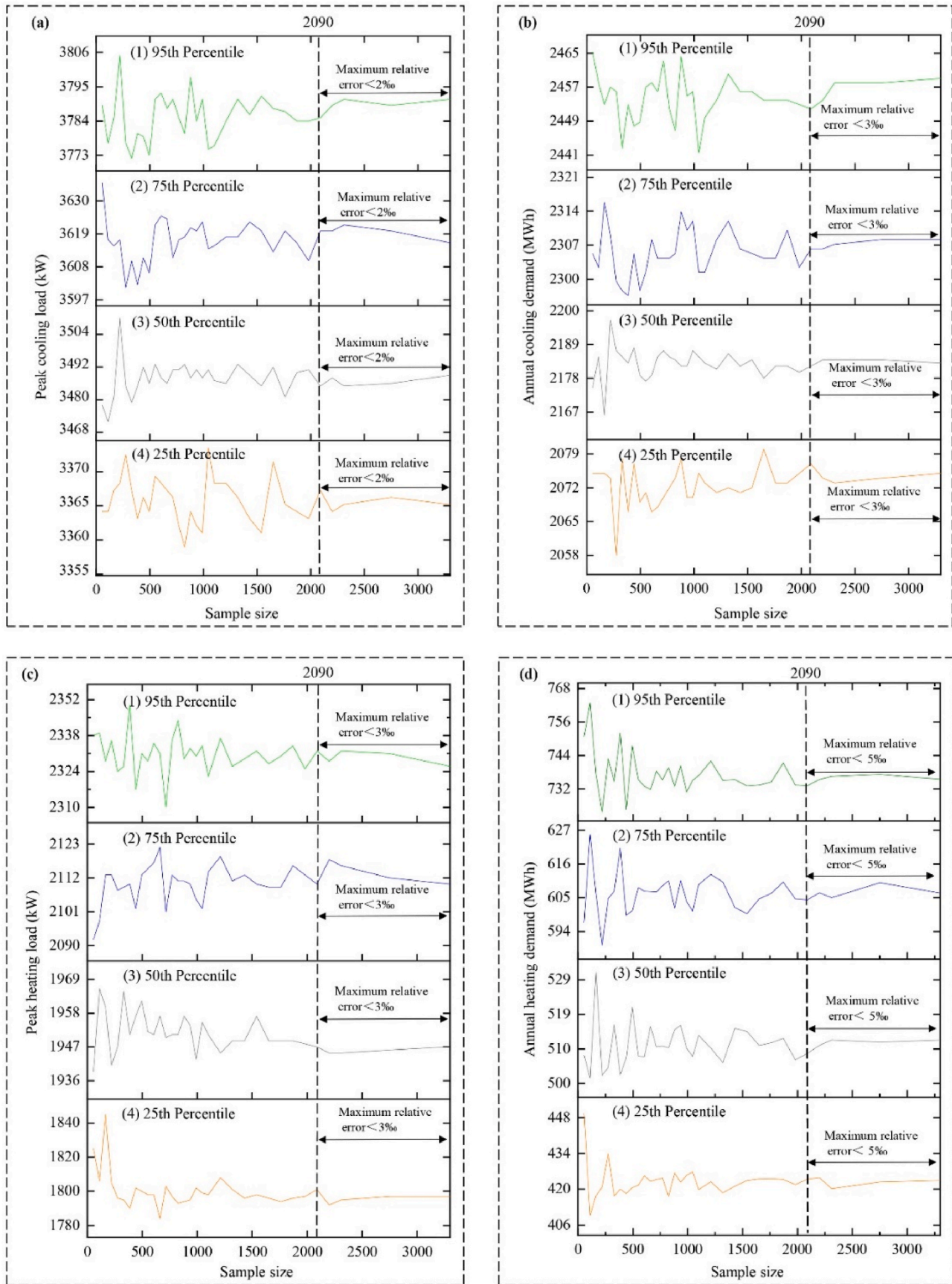


Fig. 8. The percentiles (25th, 50th, 75th, and 95th) for determining sample size of UA and GSA: (a) peak cooling load; (b) annual cooling demand; (c) peak heating load; (d) annual heating demand.

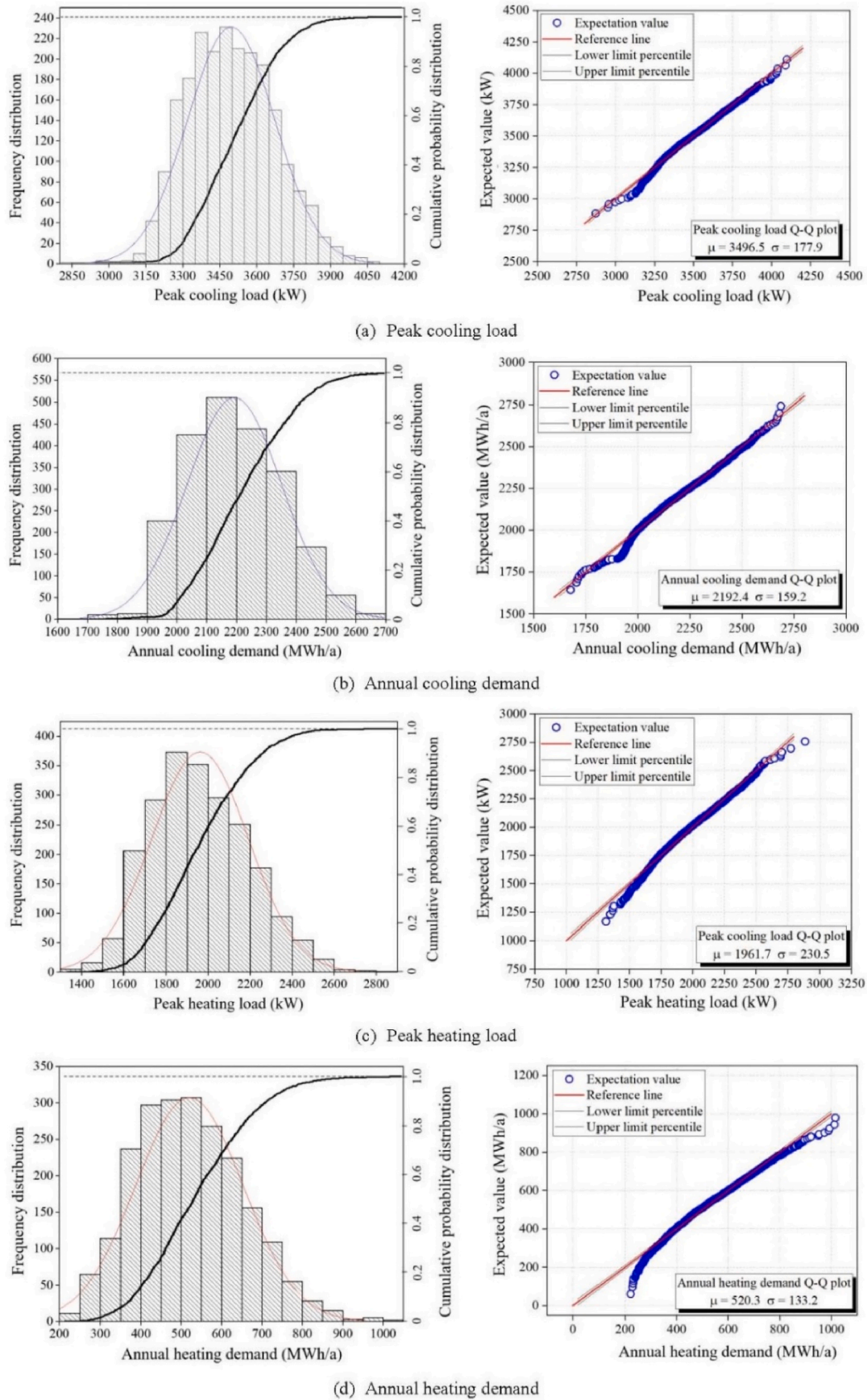


Fig. 9. The plots of frequency distribution, cumulative probability distribution and normality Q-Q: (a) peak cooling load; (b) annual cooling demand; (c) peak heating load; (d) annual heating demand.

Table 4

Mean, standard deviation, 95% interval and 95th percentile value of peak cooling load, peak heating load, annual cooling demand and annual heating demand.

Indicator	Distribution	μ (kW)	σ (kW)	95% interval (kW)	95th percentile (kW)
Peak cooling load	Normal	3496.5	177.9	[3147.8, 3845.2]	3789
Peak heating load	Normal	1961.7	230.5	[1509.9, 2413.5]	2341
Indicator	Distribution	μ (MWh)	σ (MWh)	95% interval (MWh)	95th percentile (MWh)
Annual cooling demand	Normal	2192.4	159.2	[1880.4, 2504.4]	2454
Annual heating demand	Normal	520.3	133.2	[259.2, 781.4]	739

2090 samples. Moreover, the relative errors of these four indicators after 2090 samples are less than 2‰, 3‰, 3‰ and 5‰ respectively. On the premise of meeting the accuracy requirements, reduce the sample size as much as possible to improve the work efficiency. Therefore, 2090 samples (190 times of the variables) are selected as the data basis of UA in this paper. Of course, researchers can determine different sample sizes according to the accuracy requirements of the project.

4.2. Uncertainty analysis

Fig. 9 shows the UA results of peak cooling load, annual cooling demand, peak heating load and annual heating demand based on 2090 samples. It can be seen from the Frequency distribution in Fig. 9(a) that the expected value of peak cooling load is very close to the normal distribution value. At the same time, the Q-Q plot also verifies the fact that the peak cooling load indeed obeys the normal distribution. By fitting histogram, the normal distribution with mean value (μ) of 3496.5 kW and standard deviation (σ) of 177.9 kW is calculated. In particular, it is possible to calculate the percentage of peak cooling load that lie within a band around the mean using PDF. For the normal distribution, 95% confidence interval of the peak cooling load corresponds to the interval [3147.8, 3845.2].

Much more important information about peak cooling load can be obtained. For example, the minimum, mean and maximum values of peak cooling load per unit area are 59.8 W/m², 72.8 W/m² and 85.4 W/m² respectively. If necessary, other quantile values (such as upper quartile, median and lower quartile) can also be read from the graph. What's more, any probability to exceed a threshold value can be calculated from CDF. For example, the probability to exceed 0–3789 kW for peak cooling load is 95%. At this point, a planner or decision maker could set a specific risk rely on the CDF. More specifically, planners or decision makers can plan and design the building cooling system with corresponding risk level according to the uncertain results of peak cooling load, and make full comparison and selection from the perspective of safety and economy.

Similar to the UA of peak cooling load, annual cooling demand, peak heating load and annual heating demand can also be quantified according to the needs of planners or decision makers using the plots of frequency distribution and cumulative probability distribution shown in Fig. 9 (b), Fig. 9 (c) and Fig. 9 (d). Annual cooling demand, peak heating

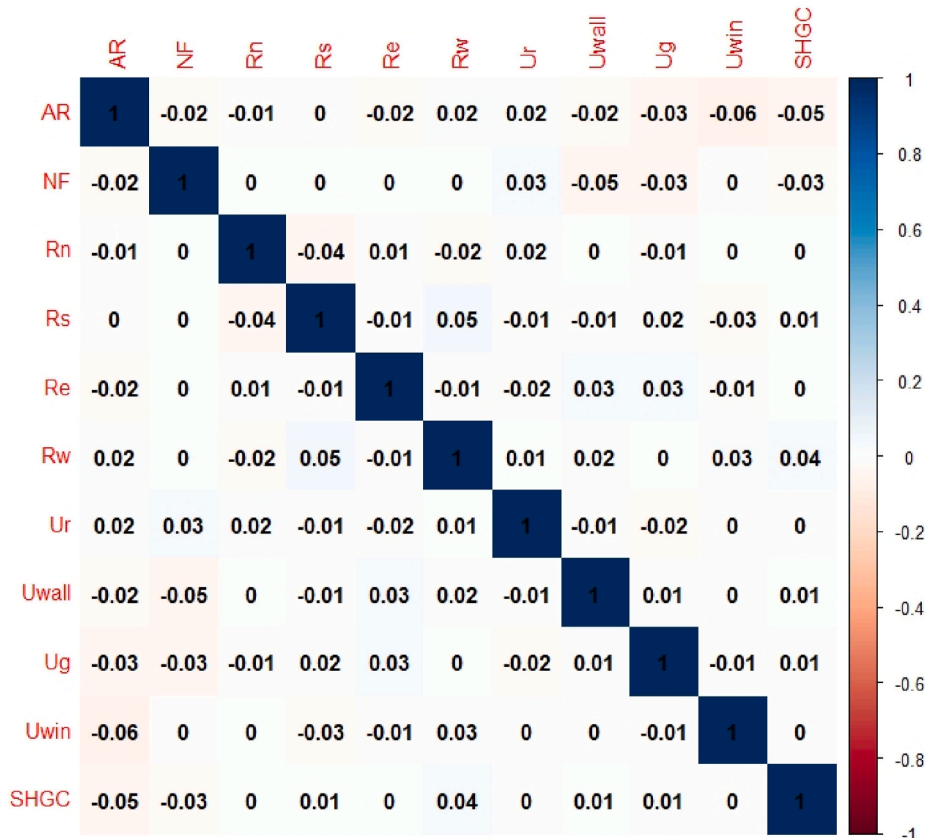


Fig. 10. Correlation coefficient between input variables.

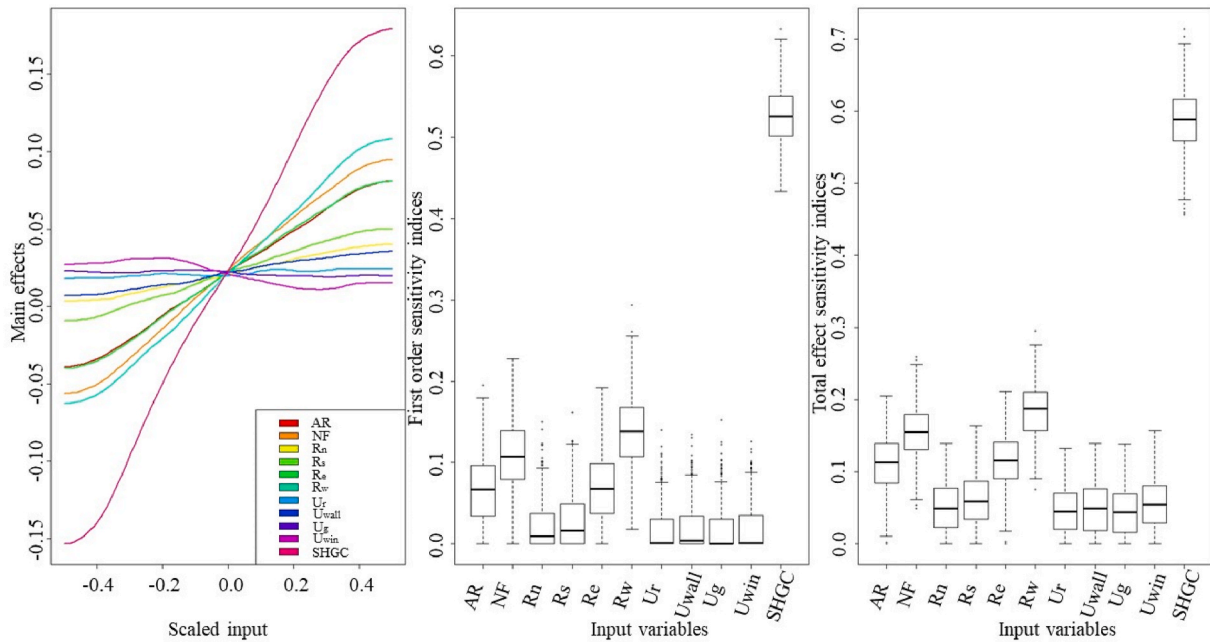


Fig. 11. GSA results of peak cooling load based on TGP.

load and annual heating demand all follow normal distributions verified by Q-Q plots, with mean values of 2192.4 MWh/a, 1961.7 kW and 520.3 MWh/a and standard deviation of 159.2 MWh/a, 230.5 kW and 133.2 MWh/a, respectively. The minimum, mean and maximum values of annual cooling demand per unit area are 34.9 kWh/(m²·a), 45.7 kWh/(m²·a) and 55.9 kWh/(m²·a) respectively, which values of peak heating load per unit area are 27.4 W/m², 40.9 W/m² and 60.1 W/m² respectively, and for annual heating demand per unit area are 4.7 kWh/(m²·a), 10.9 kWh/(m²·a) and 19.8 kWh/(m²·a) respectively. Furthermore, several important uncertainty quantification of annual cooling demand, peak heating load and annual heating demand are summarized in Table 4, including distribution function, characteristic values, confidence interval and probability to exceed the threshold value.

For buildings that need both cooling and heating, it is very crucial to compare the four indicators of peak cooling load, annual cooling demand, peak heating load and annual heating demand for the planning and design of building energy system. As can be seen from Table 4, 95th percentile value of the peak heating load is about 62% of the peak cooling load, and annual cooling demand is 3.3 times of the annual heat demand. Therefore, it is necessary to consider both cooling load and heating load for equipment configuration in order to realize the economic and efficient operation of building energy system throughout the year. Moreover, the seasonal energy storage technology and heat recovery technology also need to be considered by planners or decision makers. The UA of this case is only an example, and the uncertainty quantification of cooling and heating loads for other building energy planning need to be carried out according to the specific projects. However, the analysis method and risk decision theory can be used as a reference in other projects.

4.3. Global sensitivity analysis

This section will discuss the sensitivity results of peak cooling load, annual cooling demand, peak heating load and annual heating demand from GSA based on TGP and SRC. Before using TGP-based method and SRC method for GSA, the correlation of eleven uncertain variables needs to be analyzed. As shown in Fig. 10, the correlation coefficient between different input variables is between -0.06 and 0.05, and the absolute values of the correlation coefficient between all variables are all less than 0.1, indicating that the input variables are not correlated. In

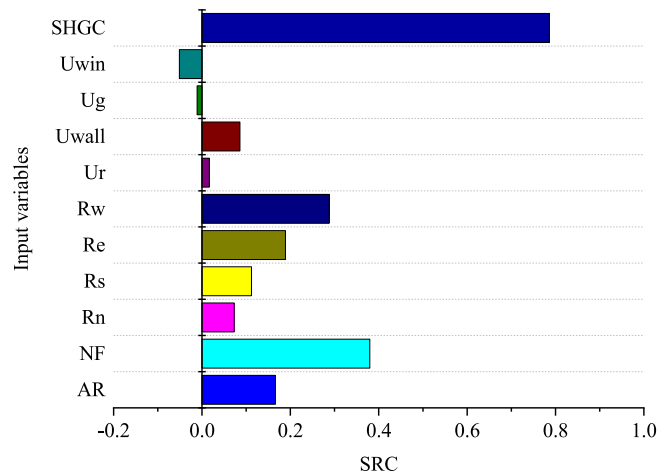


Fig. 12. GSA results of peak cooling load based on SRC.

addition, the R^2 of inputs and four outputs are 0.918, 0.889, 0.96 and 0.921 respectively. Therefore, these two methods are feasible.

As can be seen from Fig. 11, the GSA results of all the eleven variables influencing peak cooling load are obtained based on TGP. SHGC of windows is the most important factor affecting peak cooling load. This factor accounts for more than half the variations of the output for both the first order and total effects. The next two important variables are R_w and NF. In addition, R_e and AR are also important for peak cooling load. The remaining factors provide only a minor contribution to the uncertainty in peak cooling load. As shown in Fig. 12, the five most important factors affecting the peak cooling load obtained by the SRC method are the same as TGP-based method. The only difference is that R_w and NF are ranked in the opposite order in these two different methods. The influence of building floors on peak cooling load is mainly reflected in the shape coefficient of the building. The minimum number of floors in this case study is 20 and the building height is 80 m. The maximum values of the long side and short side of the building standard floor are 90 m and 50 m respectively. This leads to an overall upward trend in the shape coefficient of building with the increase of the number of floors.

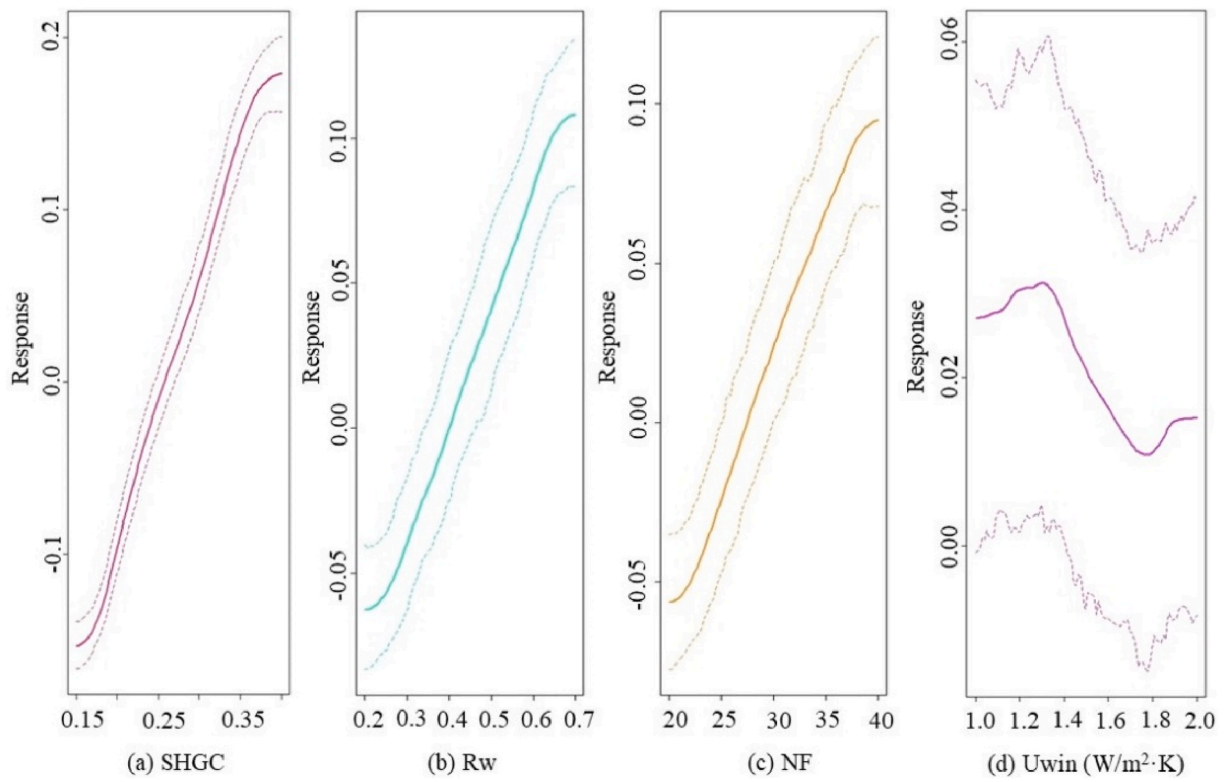


Fig. 13. Main effects of important factors for peak cooling load based on TGP: (a) SHGC; (b) R_w ; (c) NF; (d) U_{win} .

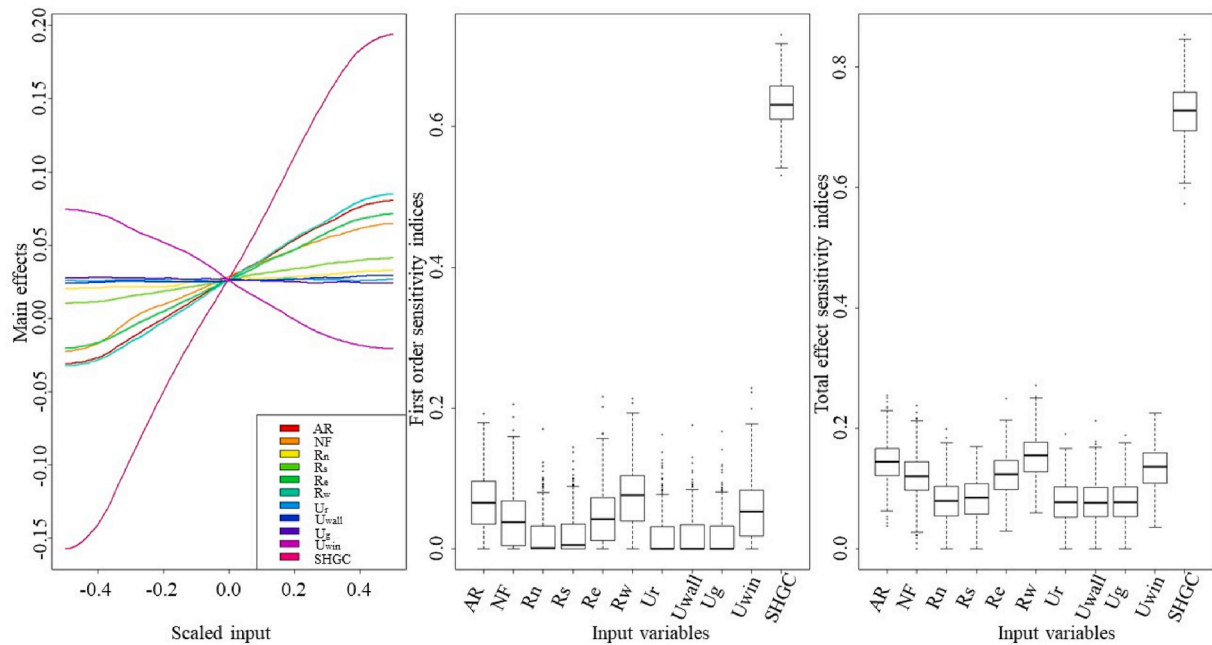


Fig. 14. GSA results of annual cooling demand based on TGP.

Therefore, the peak cooling load of the building will gradually increase.

Also can be seen from Figs. 11 and 12, except for U_{win} and U_g , the other variables have a positive correlation with the output results. Furthermore, the TGP-based method can also explain the change of peak cooling load due to the important factors. As shown in Fig. 13, the two dashed lines in this figure illustrate the 90% interval of the main effects, i.e. the lower and upper 5th percentile, respectively. As can be seen that there are linear relationships between peak cooling load and these three

inputs of SHGC, R_w and NF. Therefore, an increase of SHGC, R_w and NF will lead to higher peak cooling load. More importantly, this figure can find the key nodes of variables that change the trend of output results. For example, Fig. 13(d) clearly show that when the U_{win} is about $1.35 \text{ W/m}^2\cdot\text{K}$ or $1.8 \text{ W/m}^2\cdot\text{K}$, there is an obvious inflection point in the output curve. Compared with the SA results obtained by SRC method, this is the advantage of TGP-based method.

Similar to the above GSA of peak cooling load, the first order and

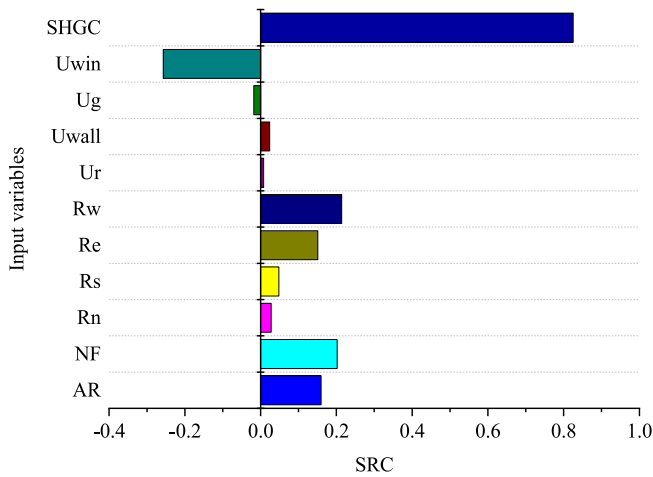


Fig. 15. GSA results of annual cooling demand based on SRC.

total effects sensitivity indices and SRC values for all the eleven input factors affecting annual cooling demand have been obtained. It can be seen from Fig. 14 that SHGC is a dominant influencing factor of annual cooling demand, which is responsible for around 70% of the variance of the output and markedly higher than any other factors in terms of both the first order and total effects sensitivity indices. The next two important factors are U_{win} and R_w . Compared with SRC results shown in Fig. 15, the first six important factors are the same, but some of them are different in order. No matter which method is used, the GSA results are that except for U_{win} and U_g , the other variables have a positive correlation with the output results. Therefore, taking effective shading measures and improving the thermal performance of windows are the main means to reduce the annual cooling demand of buildings.

As shown in Fig. 16, the first two important factors affecting peak heating load are U_{win} and NF, they account for around 60% of the output variance of first order effects and even about 70% in terms of the total effects. The next three important variables are R_e , R_w and U_{wall} . Two box plots for R_e , R_w and U_{wall} have significant overlapping, which indicates the actual effects for these three variables may be very similar. Interestingly, as shown in Fig. 16 and Fig. 17, the ranking results of factor

importance obtained by TGP-based method and SRC method are almost consistent. Moreover, the first five important factors are positively correlated with the output results. Therefore, improving the thermal performance of exterior windows and exterior walls and appropriately reducing number of floors can effectively reduce the peak heating load.

It is not difficult to see from Fig. 18 that U_{win} is the most important factor affecting annual heating demand. This factor accounts for more than 45% of the variance of the output for the first order and even close to 55% for the total effects. The next three important variables are SHGC, NF and U_{wall} , which is the same as the result obtained by SRC method shown in Fig. 19. Except for SHGC, the other variables have a positive correlation with the output results. Therefore, improving the thermal performance of exterior windows and exterior walls, appropriately reducing number of floors and shading can effectively reduce annual heating demand.

It can be seen from the results obtained by TGP-based method, the interactions between the variables are not very significant for these four indicators. The interactions between the variables are more significant for annual cooling demand compared with the other three indicators. The interaction terms can account for extra 10% of the output variances

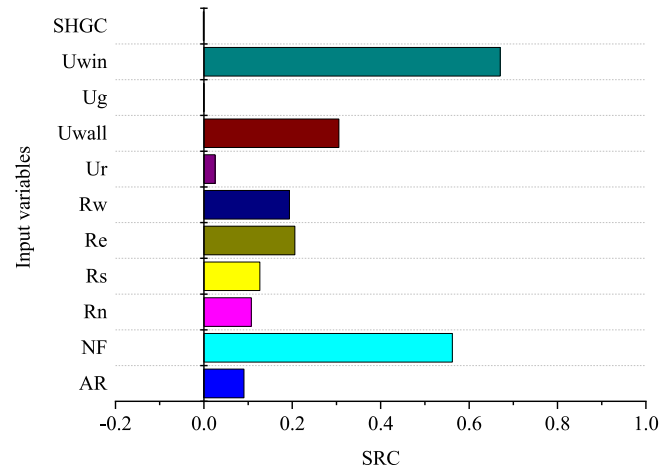


Fig. 17. GSA results of peak heating load based on SRC.

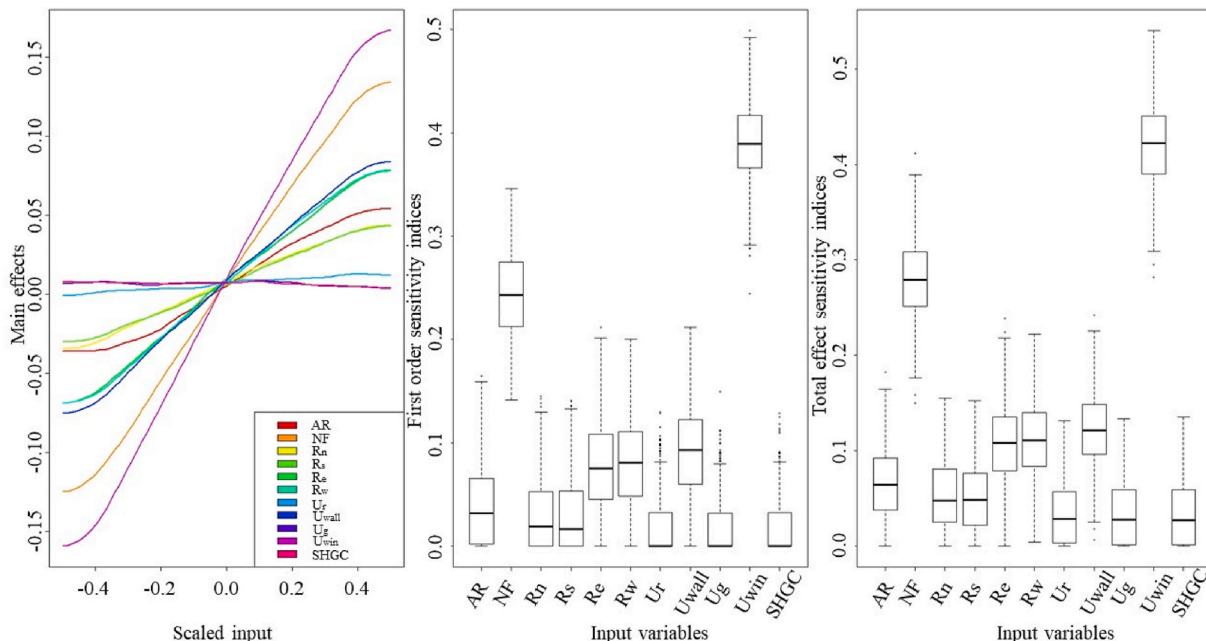


Fig. 16. GSA results of peak heating load based on TGP.

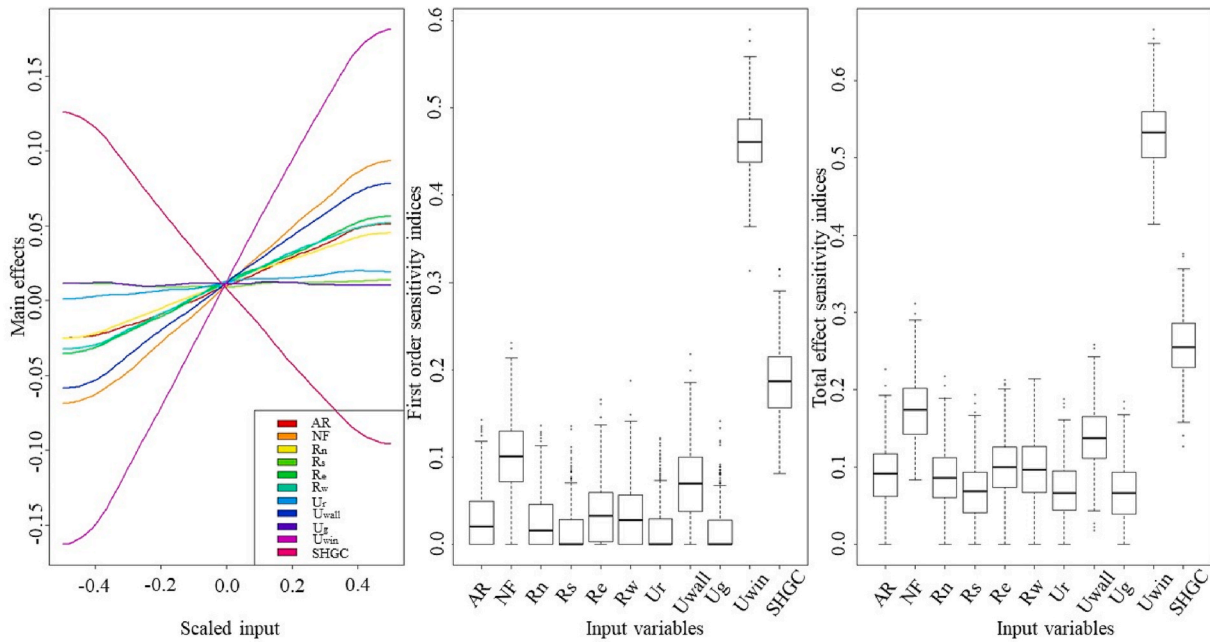


Fig. 18. GSA results of annual heating demand based on TGP.

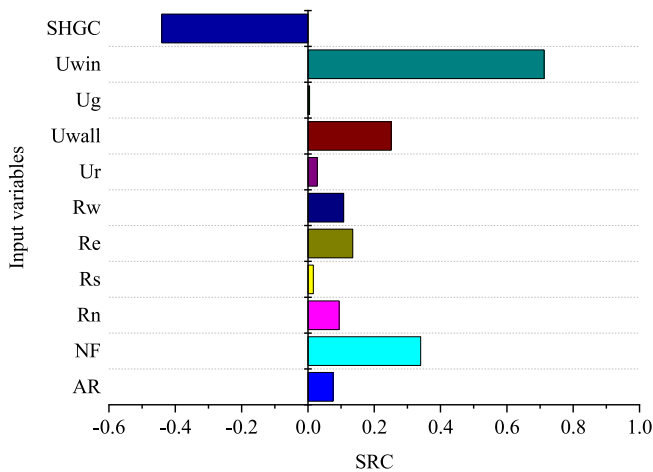


Fig. 19. GSA results of annual heating demand based on SRC.

for annual cooling demand. For the other three indicators, these interactions are only responsible for about 5% of the output variances. Furthermore, the relationships between inputs and outputs are approximately linear for all important variables. This means linear

models can be used to approximate the relationship between inputs and outputs in this case.

In summary, the five most important factors affecting these four indicators of peak cooling load, peak heating load, annual cooling demand and annual heating demand all account for more than 90% of the output variance for the first order effects, and the remaining factors only contribute minor proportions of variances for both first order and total effects. Therefore, the five most important factors and their correlations with these four indicators are summarized respectively. As can be seen from Table 5, the five most important factors affecting a single index obtained by TGP-based method and SRC method are almost the same. It shows that the SA results obtained in this study are reliable.

However, the influences of key factors on these four indicators are different. This difference can be explained in two aspects. On the one hand, the same variable has very different importance to different outputs. This means that reducing U_{win} is effective measure for decreasing peak heating load, but not necessarily significantly change peak cooling load in this building. On the other hand, the same variable is equally important for different outputs, but the effect is opposite. For example, the sensitivity indices of window SHGC for annual cooling and heating demands are all high. Obviously, reducing SHGC can decrease cooling demand but increase heating demand. Therefore, for areas requiring cooling in summer and heating in winter, reasonable shading measures should be taken to reduce cooling demand by blocking the intake of

Table 5

Five most important factors affecting peak cooling load, peak heating load, annual cooling demand and annual heating demand and their correlations with the four indicators. Note: “+” indicates that there is a positive correlation between factor and output, and “-” indicates a negative correlation between factor and output.

Peak cooling load		Peak heating load		Annual cooling demand		Annual heating demand	
TGP	SRC	TGP	SRC	TGP	SRC	TGP	SRC
SHGC (+)	SHGC (+)	U_{win} (+)	U_{win} (+)	SHGC (+)	SHGC (+)	U_{win} (+)	U_{win} (+)
R_w (+)	NF (+)	NF (+)	NF (+)	R_w (+)	U_{win} (-)	SHGC (-)	SHGC (-)
NF (+)	R_w (+)	R_e (+)	U_{wall} (+)	AR (+)	R_w (+)	NF (+)	NF (+)
R_e (+)	R_e (+)	R_w (+)	R_w (+)	U_{win} (-)	NF (+)	U_{wall} (+)	U_{wall} (+)
AR (+)	AR (+)	U_{wall} (+)	R_e (+)	R_e (+)	R_e (+)	R_e (+)	R_e (+)

Note: “+” indicates that there is a positive correlation between factor and output, and “-” indicates a negative correlation between factor and output.

solar energy in summer, and solar energy should be obtained as much as possible in winter to reduce heating demand.

According to the above GSA results, some suggestions for designers and planners can be given. On the one hand, it is easy for architectural designers to understand and master the design factors that are more important to the cooling and heating loads and the mechanism of these factors on the outputs, so as to assist the design of building energy conservation; on the other hand, energy planners can appropriately reduce the unimportant variables and focus more on important variables for the cooling and heating loads forecasting models, so as to improve the accuracy of simulation results, save the time cost of large samples and serve the practice of building energy planning with more efficient.

5. Conclusions

The deterministic building loads predictions only provide limited confidence for building energy planning. As these forecasting models involve many input parameters, the uncertainty introduced by each parameter may invalidate the entire simulation result. Therefore, the needs for uncertainty quantification and key factors identification of building loads at planning phase is quite evident. In this paper, the UA and GSA of peak cooling load, peak heating load, annual cooling demand and annual heating demand are studied in building energy planning stage. The main conclusions can be summarized as follows:

- (1) A new MC method based on BPS has been proposed to solve the problem of building loads forecasting at planning phase. The MC method constructed by the combination of R language platform and EnergyPlus software provides a new idea for rapid generation of models and accurate simulations of building loads. Note that it is necessary to assess the stability of results as a function of sample size from uncertainty analysis in applying the forecasting method into building loads assessment. This is often ignored in applying UA in the field of building loads analysis. The relative errors of the output percentiles are used as the basis for the selection of sample size in this paper.
- (2) The TGP-based GSA method is applicable to identify and analyze key variables affecting building loads. For the method, besides the point estimate of sensitivity index, the variations of these sensitivity measures can be also created to provide more robust analysis. Furthermore, the method can also show the main effects of varying each input on outputs, including the trend of outputs change due to the corresponding input. Furthermore, it is recommended to compare several SA methods and then choose at least two applicable methods that are inherently different in nature in order to provide robust analysis.
- (3) For the case study of office building in this paper, some conclusions can be drawn. Peak cooling load, annual cooling demand, peak heating load and annual heating demand all follow normal distributions. The key factors affecting above four outputs are different, including two aspects of importance ranking and positive and negative correlation. These four outputs are mainly driven by main effects and the interaction terms are not significant, and the relationships between key inputs and outputs are approximately monotonic and linear. Moreover, this study also provides insight on capacity design for building energy planning, and the energy system with large demand differences should be carefully decided based on the combination of uncertainty and sensitivity analysis.

It is worth noting that the very limited range means the impact of the building envelope on the energy demand does not change much due to over-insulation, one can be lead to believe that the U-values do not impact the results, but this is only because the variations are so small. Therefore, in order to achieve building energy conservation, we should not always blindly pursue to improve the thermal performance of

building envelope. When the thermal performance of building envelope reaches a certain level, its improvement on building energy conservation will be greatly reduced. Compared with other measures, the advantage of improving its performance may be not obvious. On the contrary, the cost will increase sharply. Therefore, the range of variables is also very important when performing sensitivity analysis. Maybe the same variables have very different effects on the results in different ranges. This is also an interesting point for future research.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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